

10(a)	<i>any nine from:</i> 1 glycolysis ; 2 glucose phosphorylated by ATP ; 3 to fructose (1,6-)bisphosphate ; 4 lysis / splits, to form 2 × triose phosphate ; 5 (triose phosphate) has hydrogen removed / is dehydrogenated / is oxidised ; 6 reduced NAD formed ; 7 4 × ATP / net 2ATP, produced ; 8 <i>ref. to</i> substrate-linked phosphorylation ; <i>pyruvate produced</i> 9 enters mitochondrial matrix ; 10 link reaction ; 11 decarboxylated / carbon dioxide removed ; 12 (pyruvate) has hydrogen removed / is dehydrogenated / is oxidised ; <i>combines with coenzyme A</i>	9
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Question	Answer	Marks
10(b)	<i>any six from:</i> 1 reduced NAD / reduced FAD, releases hydrogen ; 2 <i>ref. to</i> inner membrane / cristae ; 3 hydrogen splits into H⁺ and e⁻ ; 4 e⁻ passes along, chain of carriers / ETC ; 5 energy released used to pump H⁺ into intermembrane space ; 6 high concentration of H⁺ in intermembrane space / <i>ref.</i> proton gradient ; 7 H⁺ <u>diffuse</u> back into matrix ; 8 <i>ref. to</i> ATP synthase / stalked particle ; 9 <i>ref. to</i> chemiosmosis ;	6